Unit III: Colloidal State of Matter

Concept of colloidal and its types, Different system of colloids, Dispersed phase and dispersion medium, Methods of preparation of colloidal solutions, Dialysis and electro dialysis, Properties of colloidal solution Brownian Movement, Tyndall effect, Electrophoresis and coagulation, Protection and protective colloids.

Emulsion: Types, preparation, properties and uses, Application of colloids chemistry in different industries.

Concept of colloidal and its types

The nature of the solution formed depends upon the size of the solute particles. If the size of the solute particles is less than 1 nm it will form true solution but when the size is between 1 to 1000 nm then it will form colloidal solution. When the size of solute particles is greater than 1000 nm it will form a suspension. Therefore we may conclude that colloidal solution is an intermediate state between true solution and suspension.

S.No.	Name of Property	True Solution	Colloids Solution	Suspension
1.	Size	Size of particles is less than 1 nm	Size of particles is between 1nm and 1000 nm	Size of particles is greater than 1000 nm
2.	Filterability	Pass through ordinary filter paper and also through animal membrane.	Pass through ordinary filter paper but not through animal membrane.	Do not pass through filter paper or animal membrane.
3.	Settling	Particles do not settle down on keeping	Particles do not settle down on their own but can be made to settle down by centrifugation.	Particles settle down on their own under gravity.
4.	Visibility	Particles are invisible to the naked eye as well as under a microscope.	Particles are invisible to the naked eye but their scattering effect can be observed with the help of a microscope.	Particles are visible to the naked eye.
5.	Separation	The solute and solvent cannot be separated by ordinary filteration or by ultra filteration.	The solute and solvent cannot be separated by ordinary filteration but can be separated by ultra-filteration.	The solute and solvent can be separated by ordinary filteration.
6.	Diffusion	Diffuse quickly	Diffuse slowly	Do not diffuse

Phases of Colloids Solution

Colloids solutions are heterogenous in nature and always consist of at least two phases: the dispersed phase and the dispersion medium.

- **Dispersed Phase:** It is the substance present in small proportion and consists of particles of colloids size (1 to 1000 nm). Such particles are normally easily visible in an optical microscope.
- **Dispersion Medium:** It is the medium in which the colloids particles are dispersed.

The colloidal state is very widespread in biological systems and in many practical situations. Unlike solutions, colloids do not constitute a solute dissolved in the solvent phase. Rather, the solute phase is dispersed in the solvent phase. For example, in a colloidal solution of sulphur in water, sulphur particles constitute the 'dispersed phase' and water is the 'dispersion medium'. Each of the two phases namely, dispersed phase and dispersion medium can be solid, liquid or gas. Thus, different types of colloidal solutions are possible depending upon the physical state of the two phases. For example:

Sol – It is a suspension of minute solid particles in a liquid.

Emulsion – It is a colloid between two or more liquid with one consisting a dispersion of another liquid.

Foam – It consists of gas dispersed in solid or liquid.

Aerosol – It consists of minute liquid or solid particles in a gas.

Different types of colloidal solutions and their examples are shown in Table:

Dispersed phase	Dispersion medium	Name	Examples
Solid	Solid	Solid-sol	Ruby glass, alloys
Solid	Liquid	Sol	paint, ink
Solid	Gas	Aerosol	Smoke, haze
Liquid	Solid	Gel	Curd, cheese
Liquid	Liquid	Emulsion	Milk, Cream
Liquid	Gas	Liquid Aerosol	Cloud, mist
Gas	Solid	Solid form	Cork, pumice stone
Gas	Liquid	Foam	Froths of air

Note:

- Gases cannot form a colloidal solution between themselves, because they form homogenous mixtures.

- Out of the various types of colloidal solutions listed above, the most common are sols (solid in liquid type), gels (liquid in solid type) and emulsions (liquid in liquid type). If the dispersion medium is water then the 'sol' is called a hydrosol; and if the dispersion medium is alcohol then the 'sol' is called an alcosol.

Classification

Depending upon the interaction between dispersed phase and the dispersion medium colloidal solutions have been classified into two categories.

1. **Lyophilic colloids:** The word Lyophilic means solvent lover. Lyophilic colloidal solutions are those in which the dispersed phase has a great affinity (or love) for the dispersion medium. Substances like gum, gelatine, starch etc. when mixed with suitable dispersion medium, directly pass into colloidal state and form colloidal solution. Therefore, such solutions are easily formed simply by bringing dispersed phase and dispersion medium in direct contact with each other. However, these colloidal solutions have an important property i.e. they are **reversible** in nature. This means that once lyophilic colloidal solution has been formed then dispersed phase and dispersion medium can be separated easily. Once separated these can again be formed by remixing the two phases. These sols are quite stable. If water is used as dispersion medium then it is termed as hydrophilic colloid.
2. **Lyophobic Colloids:** The word Lyophobic means solvent hating. Lyophobic colloidal solutions are those in which the dispersed phase has no affinity for the dispersion medium. Metals like Au, Ag and their hydroxides or sulphides etc., when simply mixed with dispersion medium do not pass directly into colloidal state. These sols have to be prepared by special methods. These sols can be readily precipitated and once precipitated they have little tendency to go back into the colloidal state. Thus these sols are **irreversible** in nature. Also they are not very stable and require a stabilizing agent to remain in the colloidal form. In case water is used as dispersion medium it is called as hydrophobic sol.

Methods of preparation of colloidal solutions

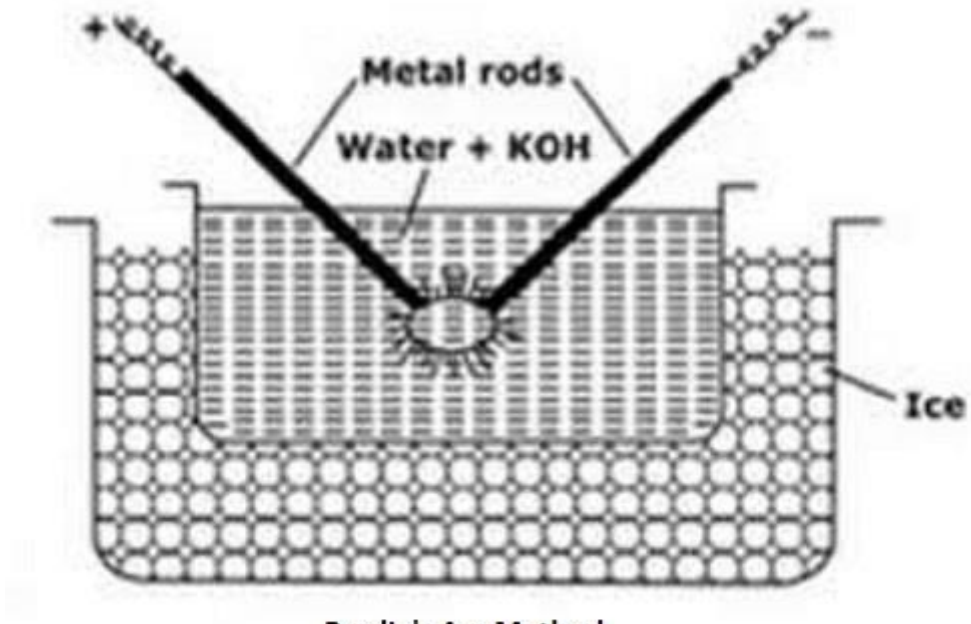
The lyophilic sols can be prepared directly by mixing the dispersed phase with the dispersion medium. For example, colloidal solutions of starch, gelatin, gum etc. are prepared by simply dissolving these substances in hot water. Similarly, a colloidal sol of cellulose nitrate is obtained by dissolving it in alcohol. The resulting solution is called collodion.

However, lyophobic colloids cannot be prepared by direct method. Hence two types of methods are used for preparing lyophobic colloids. These are:

- Physical methods
- Chemical methods

Physical methods

- **Bredig's Arc Method:** These methods are employed for obtaining colloidal solutions of metals like gold, silver, platinum etc.**.** An electric arc is struck between the two metallic electrodes placed in a container of water. The intense heat of the arc converts the metal into vapours, which are condensed immediately in the cold water bath. This results in the formation of particles of colloidal size. We call it as gold sol.



Bredig's Arc Method

- **Peptisation:** Peptisation is the process of converting a freshly prepared precipitate into colloidal form by the addition of a suitable electrolyte. The electrolyte is called peptising agent.
  For example when ferric chloride is added to a precipitate of ferric hydroxide, ferric hydroxide gets converted into reddish brown coloured colloidal solution. This is due to preferential adsorption of cations of the electrolyte by the precipitate. When $FeCl_3$ is added to $Fe(OH)_3$, Fe^{3+} ions from $FeCl_3$ are adsorbed by $Fe(OH)_3$ particles. Thus the $Fe(OH)_3$ particles acquire + ve charge and they start repelling each other forming a colloidal solution.

Chemical Methods: By oxidation

- Sulphur sol is obtained by bubbling H_2S gas through the solution of an oxidizing agent like HNO_3 or Br_2 water, etc. according to the following equation **:**

$$Br_2 + H_2S \rightarrow S + 2\,HBr$$

$$2\,HNO_3 + H_2S \rightarrow 2\,H_2O + 2\,NO_2 + S$$

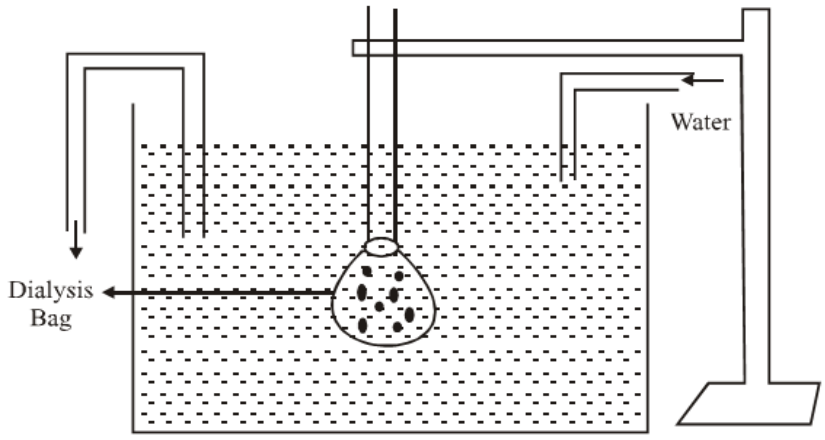
$Fe(OH)_3$ sol, As_2S_3 sol can also be prepared by chemical methods.

Purification of Colloidal Solution

When a colloidal solution is prepared it contains certain impurities. These impurities are mainly electrolytic in nature and they tend to destabilise the colloidal solutions. Therefore colloidal solutions are purified by the following methods:

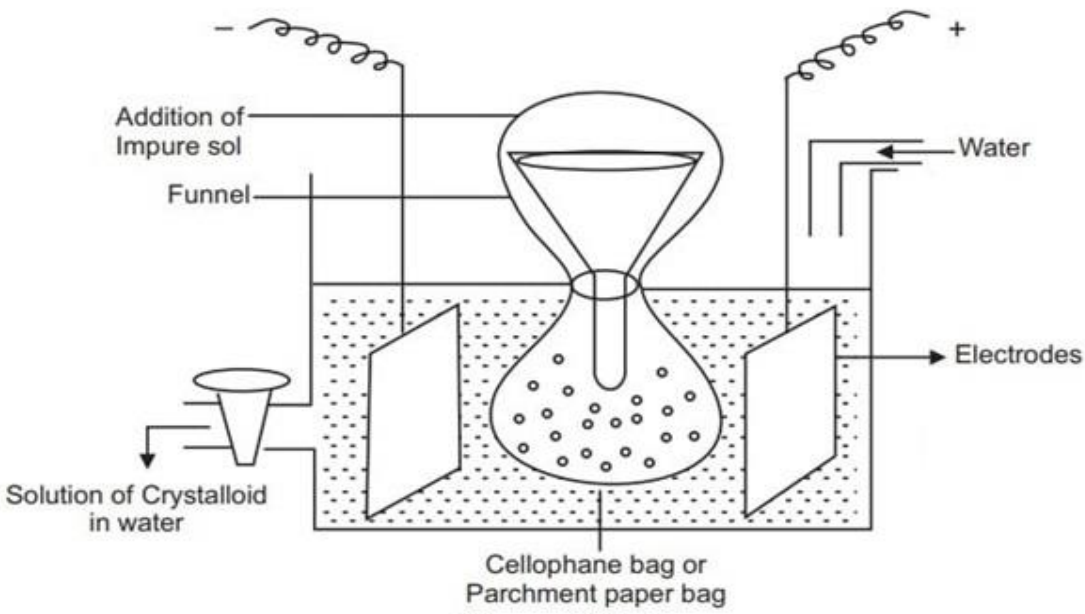
- Dialysis
- Electrodialysis

Dialysis: The process of dialysis is based on the fact that colloidal particles cannot pass through parchment or celloplane membrane while the ions of the electrolyte can. The colloidal solution is taken in a bag of cellophane which is suspended in a tub full of fresh water. The impurities diffuse out leaving pure colloidal solution in the bag. This process of separating the particles of colloids from impurities by means of diffusion through a suitable membrane is called dialysis.



Dialysis

Electrodialysis: The dialysis process is slow and to speed up its rate, it is carried out in the presence of an electrical field. When the electric field is applied through the electrodes, the ions of the electrolyte present as impurity diffuse towards oppositely charged electrodes at a fast rate. The dialysis carried out in the presence of electric field is known as electrodialysis.



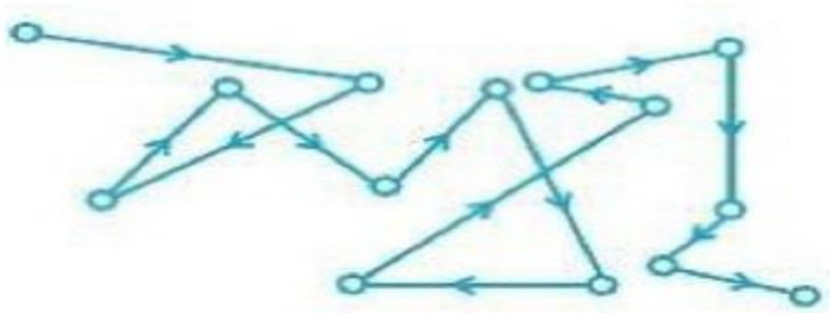
Electrodialysis

The most important use of dialysis is the purification of blood in the artificial kidney machine. The dialysis membrane allows the small particles (ions etc.) to pass through, whereas large size particles like haemoglobin do not pass through the membrane.

Properties of Colloids

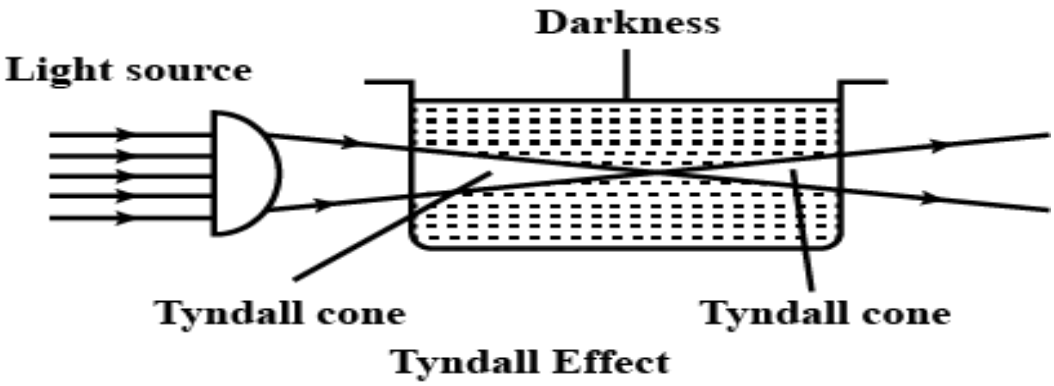
The properties of colloids are discussed below:

- **Heterogeneous character:** Colloidal particles remain within their own boundary surfaces which separate them from the dispersion medium. So a colloidal system is a heterogeneous mixture of two phases. The two phases are dispersed phase and dispersion medium.
- **Brownian movement:** It is also termed as Brownian motion and is named after its discoverer Robert Brown (a Botanist.) Brownian Motion is the zig-zag movement of colloidal particles in continuous random manner. Brownian motion arises because of the impact of the molecules of the dispersion medium on the particles of dispersed phase. The forces are unequal in different directions. Hence it causes the particles to move in a zig-zag way.



Brownian Movement

- **Tyndall Effect:** Tyndall in 1869, observed that if a strong beam of light is passed through a colloidal solution then the path of light is illuminated. This phenomenon is called Tyndall Effect. This phenomenon is due to scattering of light by colloidal particles. The same effect is noticed when a beam of light enters a dark room through a slit and becomes visible. This happens due to the scattering of light by particles of dust in the air.



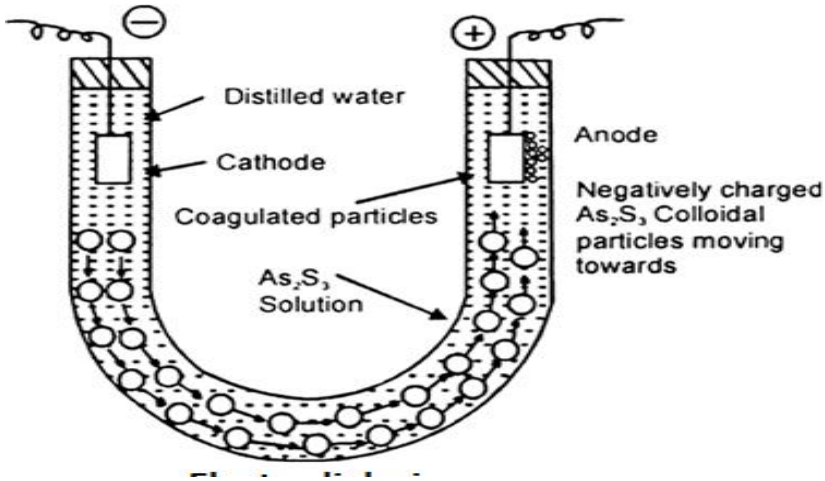
- **Electrophoresis:** The particles of a colloidal solution are electrically charged and carry the same type of charge, either negative or positive. The dispersion medium has an equal and opposite charge. The colloidal particles therefore repel each other and do not cluster together to settle down. For example, arsenious sulphide sol, gold sol, silver sol, etc. contain negatively charged colloidal particles whereas ferric hydroxide, aluminium hydroxide etc. contain positively charged colloidal particles. Origin of charge on colloidal particles is due to:

  (a) Preferential adsorption of cations or anions by colloidal particles.

  (b) Miscelles carry a charge on them.

  (c) During the formation of colloids especially by Bredig arc method, colloidal particles capture electrons and get charged. The existence of charge on a colloidal particle is shown by a process called electrophoresis.

  Electrophoresis is a process which involves the movement of colloidal particles either towards cathode or anode under the influence of electrical field.



- **Coagulation:** Coagulation is the process of aggregating colloidal particles into an insoluble precipitate by adding an appropriate electrolyte.

The presence of tiny amounts of suitable electrolytes is required for colloidal stability. When a higher concentration of an electrolyte is applied, the particles of the sol take up the oppositely charged ions and neutralise them. The neutral particles subsequently begin to aggregate, resulting in bigger particles that are precipitated. The aggregation of particles at lower concentrations of electrolytes is called flocculation, which may be reversed by shaking, but coagulation occurs at greater concentrations of electrolytes and cannot be reversed by shaking. Colloidal particles contain charges and this contributes to the stability of lyophobic colloids. If the charge is eliminated in any way, the particles will cluster together to form aggregates and settle down under gravity's influence.

Coagulation of Lyophobic Sols

1. **By electrophoresis:** The charged colloidal particles migrate towards the oppositely charged electrodes, discharge and precipitate.
2. **Coagulation by mixing sols that are oppositely charged:** When sols that are oppositely charged are mixed in appropriate amounts, they precipitate by neutralising the charges completely or partially. This type of coagulation is known as mutual coagulation.
   Example: Mixing hydrated ferric oxide (positive sol) and arsenious sulphide (negative sol) precipitates them.
3. Coagulation by boiling: Boiling a sol causes the dispersion medium molecules to collide more, which disturbs the adsorbed layer. Increased collisions result in a reduction in particle charge, which causes particles to settle as a precipitate.
4. **Coagulation by prolonged dialysis:** If the electrolyte present in the sol is fully eliminated during continuous dialysis, the colloid becomes unstable and eventually coagulates.
5. **Coagulation by the addition of electrolytes:** When an excess of an electrolyte is added to the colloidal solution, the colloidal particles get precipitated. Colloid particles neutralise by interacting with oppositely charged ions, known as coagulating

(flocculating) ions which leads to their coagulation. A positive ion causes the precipitation of negatively charged sol and vice versa.

Hardy-Schulze Rule: The following findings were drawn from experimental observations of the coagulation of various sols with various electrolytes.

1. Coagulation can be caused by ions whose charge is the opposite of that of the colloidal particles.
2. The greater the valency of the flocculating ion, the lesser is the amount of electrolyte required to coagulate a given amount of sol at a given time.

The above generalisations are known as the Hardy-Schulze rule.

As the valency of the added flocculating ion increases, its coagulation power increases. This power of flocculating ions to coagulate the colloidal sol is known as the coagulating (or flocculating) power.

For positively charged sols, the order of coagulating power of anions (coagulating ions) is

$[Fe(CN)_6]^{4-} > PO_4^{3-} > SO_4^{2-} > Cl^-$

For negatively charged sols, the order of coagulating power of cations (coagulating ions) is

$Al^{3+} > Ba^{2+} > Na^+$

Flocculating Value: Coagulation of a colloidal solution by an electrolyte does not take place until the added electrolyte has a certain minimum concentration in the solution. The flocculating value is the lowest amount of an electrolyte, measured in millimoles, needed to cause coagulation in one litre of colloidal solution.

Coagulation of Lyophilic Sols

Coagulation of Lyophilic Sols can be done by following two methods:

1. By adding an electrolyte: When an excess electrolyte is added to the colloidal sol, the colloidal particles are precipitated.
2. By adding suitable solvent.

Example: Alcohol and acetone are added to hydrophilic sol. Dehydration of the dispersion phase occurs when alcohol and acetone are added to the hydrophilic sol.

Protection and protective colloids

Since lyophobic sols are less stable than lyophilic sols, lyophilic sols have a unique property to protect lyophobic sols. Lyophilic sols used for this purpose are known as protective colloids. When lyophilic sol is added to lyophobic sol, lyophobic particles are surrounded by a layer of hydrophilic particles. Lyophilic particles protect lyophobic sol from electrolytes. In other words, on mixing both types of colloids, lyophilic colloid particles surround lyophobic colloid particles and thus lyophobic colloids are protected by the lyophilic sols.

Example: Gelatin (lyophilic) protects gold sol (lyophobic)

- **Gold Number**

The minimum weight in milligrams of a protective colloid required to prevent the coagulation of 10 mL of a standard gold sol when 1 mL of a 10% NaCl solution is added to the sol is called the Gold number of the lyophilic sol.

The smaller the gold number, the greater will be protecting power of the protective colloid.

Protecting power of protective colloid $\propto$ Gold number

Emulsion

Emulsions are colloidal solutions in which both the dispersed phase and dispersion medium are liquids. However, the two liquids are immiscible, as miscible liquids will form true solution. Emulsions are of two kinds:

(a) **Oil-in-water emulsion:** Here the dispersed phase is oil while the dispersion medium is water. Milk is an example of this kind as in milk liquid fats are dispersed in water. Vanishing cream is another example.
(b) **Water-in-oil emulsion:** Here dispersed phase is water and dispersion medium is oil. Butter, cod- liver oil, cold creams are examples of this type.

The liquids forming emulsion i.e. oil and water will separate out on keeping as they are immiscible. Therefore an emulsifying agent or emulsifier is added to stabilise the emulsion. Soap is a common emulsifier. The preparation of emulsion in the presence of an emulsifier is called emulsification.

It is believed that an emulsifier gets concentrated at the interface between oil and water i.e. the surface at which oil and water come in contact with each other. It acts as a binder between oil and water.

Applications of Emulsions - Emulsions play very important role in our daily life. Some of the common applications are given below:

1. The cleansing action of soap and synthetic detergents for washing clothes, bathing etc. is based upon the formation of oil in water type emulsion.
2. Milk is an emulsion of fat in water. Milk cream and butter are also emulsions.

3. Various cold creams, vanishing creams, body lotions etc. are all emulsions.
4. Various oily drugs such as cod liver oil are administered in the form of emulsion for their better and faster absorption. Some ointments are also in the form of emulsions.
5. The digestion of fats in the intestine occurs by the process of emulsification.
6. Emulsions are used for concentrating the sulphide ores by froth flotation process. Finely powdered ore is treated with an oil emulsion and the mixture is vigorously agitated by compressed air when the ore particles are carried to the surface and removed.

Application of colloids chemistry in different industries

1) Pharmaceuticals- Colloidal medications are more effective since they are readily absorbed by the body's tissues.

2) Treatment of wastewater- Colloidal particles of filth, muck, and other organic materials have an electric charge when sewage water is passed over the plates with high potential, the colloidal particles coagulate, and the suspended matter is removed.

3) Purification of water- It is possible to precipitate colloidal pollutants in water by adding certain electrolytes such as alum. Contaminant colloidal particles that are negatively charged are neutralised by the Al3+ ions and settle to the bottom, allowing pure water to be decanted.

4) The cleaning activity of soap- Soap solution that is colloidal in nature. It removes dirt particles by adsorbing or emulsifying the oily substance stuck to the cloth.

5) Formation of the Delta- Colloidal particles of clay, sand and a range of other things may be found in river water. Seawater has a large number of dissolved electrolytes. The colloidal particles in river water coagulate and settle at the contact site when they come into contact with salt water. As a consequence, the level of the riverbed increases. As a consequence, the water flows in a different direction, ultimately forming a delta.

6) Electrical precipitation of smoke- Smoke particles are electrically charged colloidal carbon particles in the air. To precipitate smoke particles, the Cottrell precipitator, which is based on the electrophoresis idea, is utilised. Smoke is allowed to travel through a chamber made up of a succession of metal plates linked to a high-potential source via a metal wire. The heated air escapes via the chimney when charged particles of smoke are attracted to an electrode with an opposing charge. Furthermore, dust particles are removed during this procedure. As a result, the nuisance of smoke in huge industrial cities may be avoided.

7) Techniques of photography- To make sensitive plates in photography, a colloidal solution of silver bromide in gelatin is applied to glass plates, celluloid films, or paper.

8) Rain created by humans- Spraying colloidal dust or sand particles with opposite charges over a cloud may produce artificial rain. Colloidal water droplets in the cloud will be neutralised and coagulated into bigger water drops, creating artificial rain.

9) Rubber production- Negatively charged rubber particles suspended in a colloidal fluid make up latex. Latex coagulation may be used to produce rubber. Rubber-plated items are made by depositing negatively charged rubber particles onto the object to be rubber-plated and then employing the item as an anode in a rubber plating solution.

10) Smokes-screen- Smoke screens are used to hide things with smoke clouding. Generally used to cover the movement of troops. A smoke screen is also a colloidal system in which titanium oxide particles are dispersed into the air.